

Software Development Lifecycles

Introduction

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Learning Outcomes

LO1

- Describe different software development lifecycles.

LO2

- Explain the importance of a feasibility study.

LO3

- Undertake a software development lifecycle.

LO4

- Discuss the suitability of software behavioural design techniques.

Why we need SDLC ?

1. For better software quality,
2. Make the goals of the software clear to the team (developers know what to build and testers know what and how to test)
3. Reduce the rate of vulnerabilities (fewer or none)
4. Management control.
5. Effective documentation and reduced dependencies
6. Effective resource utilization.
7. Effective cost and time definition.
8. Ensure the architecture and design are secure.
9. Meet and exceed Customer's expectations.

Software Development life cycle (SDLC)

- SDLC is a process used by the software industry to design, develop and test the quality of software.
- The SDLC aims to produce a high-quality software that meets or exceeds customer expectations, reaches completion within times and cost estimates.
- The SDLC defines a methodology for improving the quality of software and the overall development process.

Software Development life cycle (SDLC) (cont.)

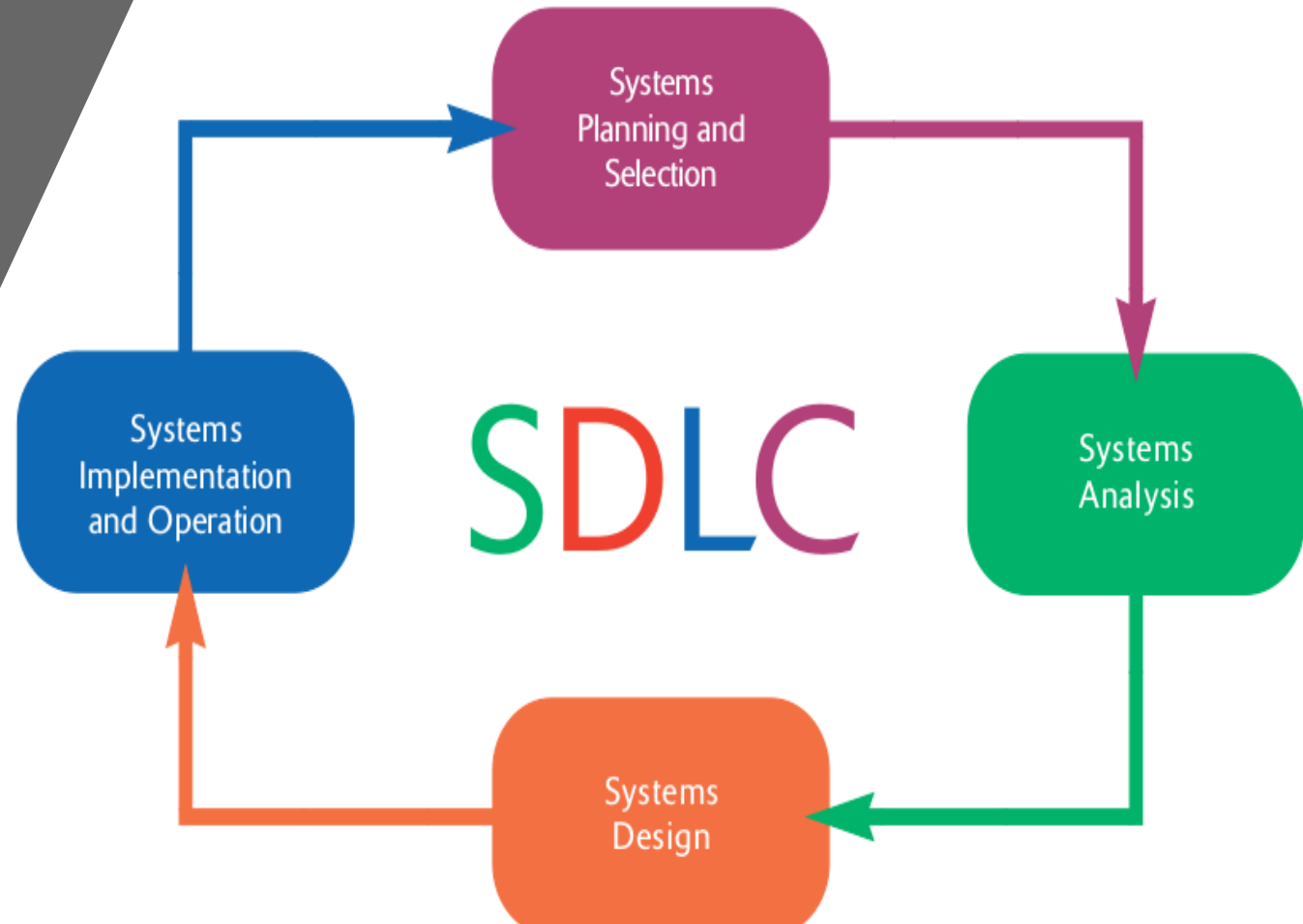
building an information system is similar to building a house.

- First, the owner describes the vision for the house to the developer.
- Second, this idea is transformed into sketches and drawings that are shown to the owner and refined (often, through several drawings, each improving on the other) until the owner agrees that the pictures depict what he or she wants.
- Third, a set of detailed blueprints is developed that presents much more specific information about the house



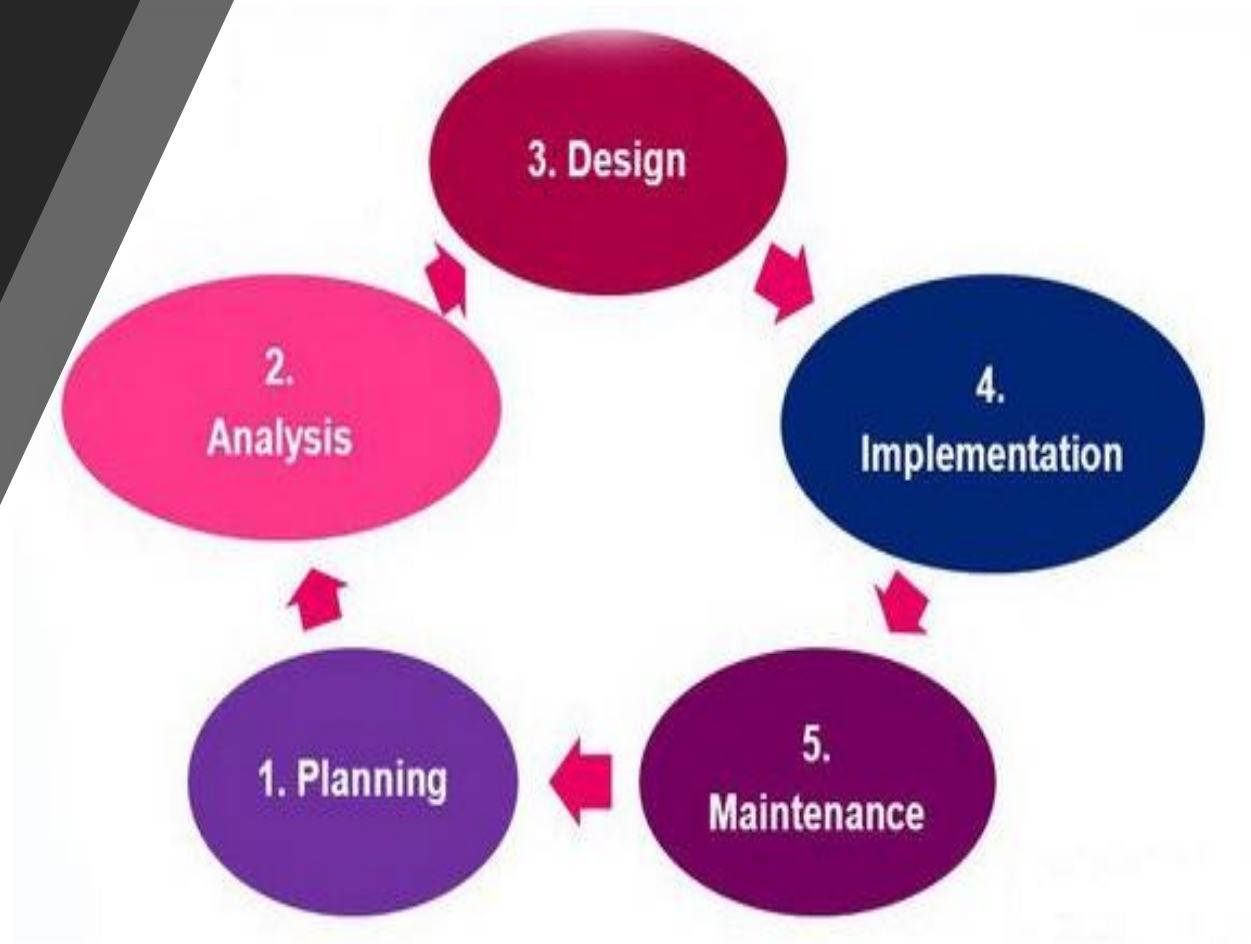
SDLC Stages

Four stages SDLC



SDLC Stages

Five stages SDLC



SDLC Stages

six stages SDLC



SDLC stages

Seven stages of SDLC

PLANNING

ANALYSIS

DESIGN

DEVELOPMENT

TESTING

DEPLOYMENT

MAINTENANCE



Planning

- Why should we build this system?

Project planning

- is a formal document designed to guide the control and execution of a project. A project plan is the key to a successful project and is the most important document that needs to be created when starting any business project.

Understanding

- Why an information system should be developed
- Creating a plan for how the project team will develop it.
- deliver a project plan.

Feasibility Study

Technical Feasibility: Can We Build It?

- Familiarity with application: Less familiarity generates more risk.
- Familiarity with technology: Less familiarity generates more risk.
- Project size: Large projects have more risk.
- Compatibility: The harder it is to integrate the system with the company's existing technology, the higher the risk will be.

Economic Feasibility (cost-benefit analysis): Should We Build It?

- Development costs
- Annual operating costs
- Annual benefits (cost savings and/or increased revenues)
- Intangible benefits and costs

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Organizational Feasibility: If We Build It, Will They Come?

- Project champion(s)
- Senior management
- Users
- Other stakeholders
- Is the project strategically aligned with the business?

Analysis

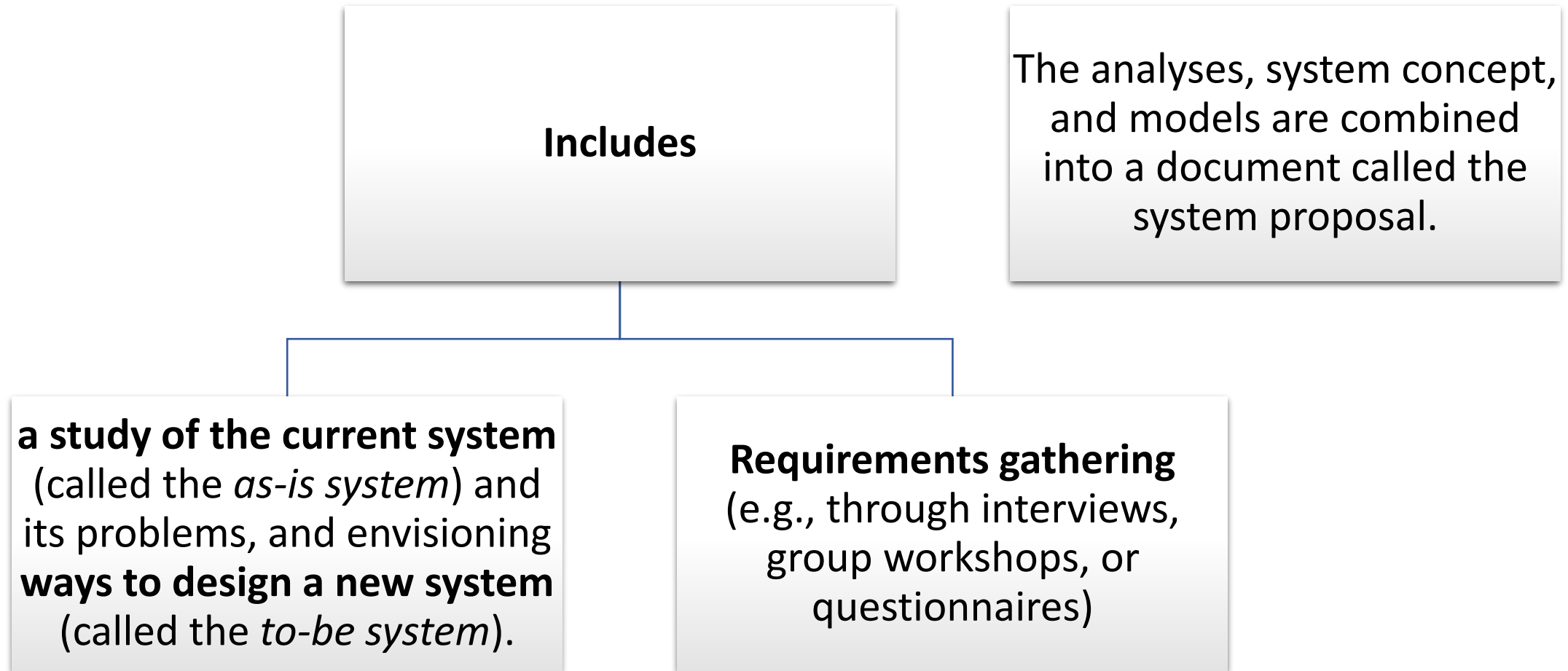
who will use the system, what the system will do, and where and when it will be used?

Develop analysis strategy

Determine Business Requirements

Output: System Proposal

Analysis



System Proposal

The system proposal is the initial deliverable that describes what business requirements the new system should meet, Which is presented to the project sponsor and other key decision makers (e.g., members of the approval committee) who will decide whether the project should continue to move forward.

Design

- The design phase decides how the system will operate in terms of the hardware, software, and network infrastructure that will be in place; the user interface, forms, and reports that will be used; and the specific programs, databases, and files that will be needed.
- Output : System Specification



Design

There are two kinds of design documents developed in this phase:

High-Level Design (HLD)

- Brief description and name of each module
- An outline about the functionality of every module
- Interface relationship and dependencies between modules
- Database tables identified along with their key elements
- Complete architecture diagrams along with technology details

Low-Level Design (LLD)

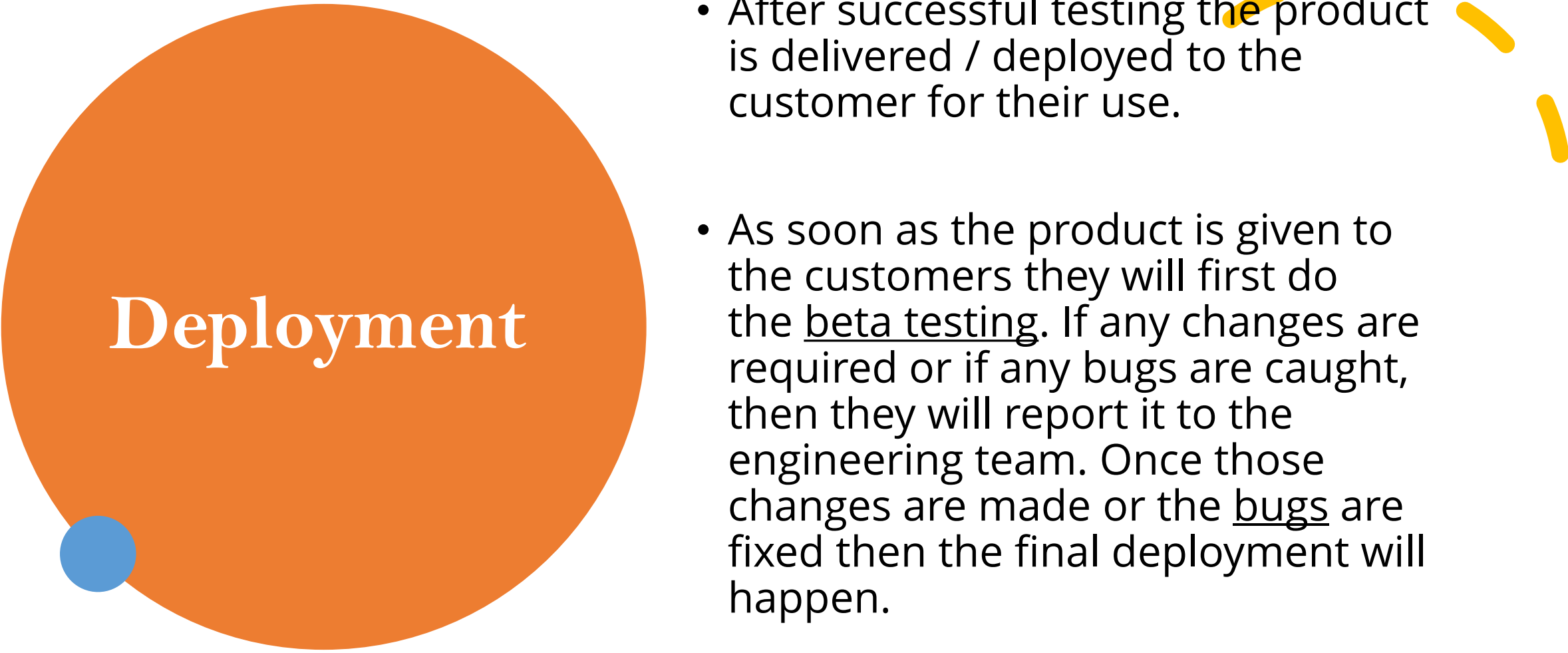
- Functional logic of the modules
- Database tables, which include type and size
- Complete detail of the interface
- Addresses all types of dependency issues
- Listing of error messages
- Complete input and outputs for every module

Implementation or coding

- Once the system design phase is over, the next phase is coding. In this phase, developers start build the entire system by writing code using the chosen programming language. In the coding phase, tasks are divided into units or modules and assigned to the various developers. It is the longest phase of the Software Development Life Cycle process.
- In this phase, Developer needs to follow certain predefined coding guidelines. They also need to use programming tools like compiler, interpreters, debugger to generate and implement the code.

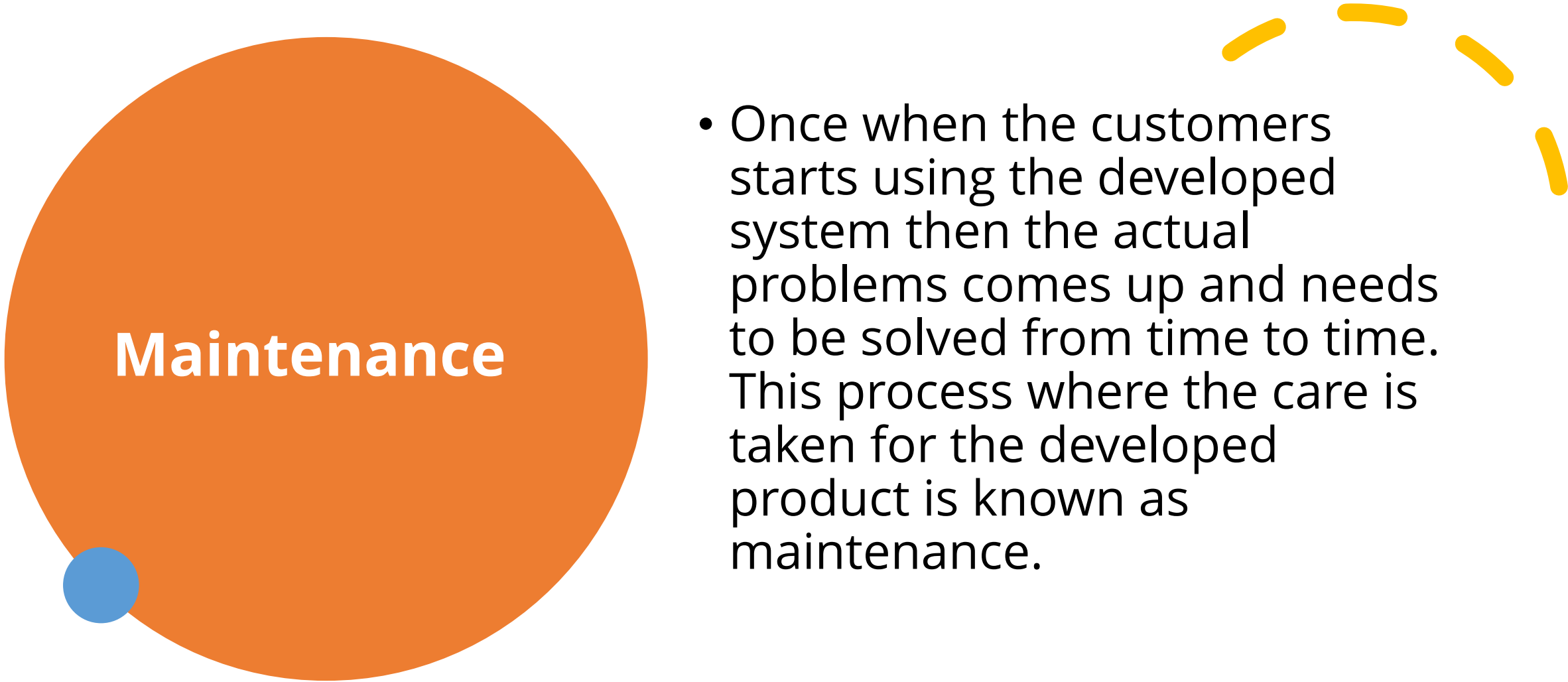
Testing

- Once the software is complete, and it is deployed in the testing environment. The testing team starts testing the functionality of the entire system. This is done to verify that the entire application works according to the customer requirement.
- During this phase, QA and testing team may find some bugs/defects which they communicate to developers. The development team fixes the bug and send back to QA for a re-test. This process continues until the software is bug-free, stable, and working according to the business needs of that system.
- During this phase all types of functional testing like unit testing, integration testing, system testing, acceptance testing are done .



Deployment

- After successful testing the product is delivered / deployed to the customer for their use.
- As soon as the product is given to the customers they will first do the beta testing. If any changes are required or if any bugs are caught, then they will report it to the engineering team. Once those changes are made or the bugs are fixed then the final deployment will happen.



Maintenance

- Once when the customers starts using the developed system then the actual problems comes up and needs to be solved from time to time. This process where the care is taken for the developed product is known as maintenance.

THE END

- <https://www.youtube.com/watch?v=i-QyW8D3ei0>